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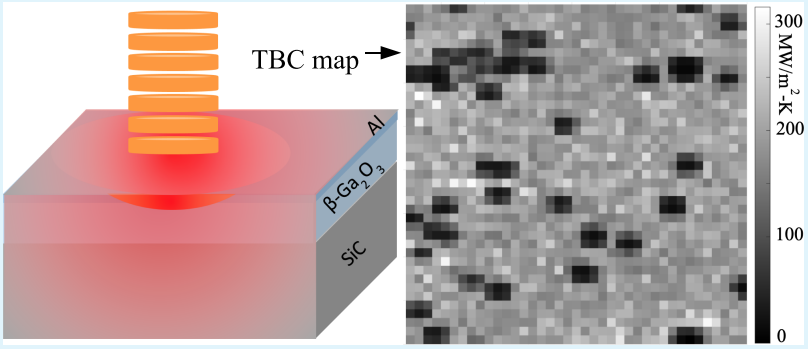
Thermal Visualization of Buried Interfaces Enabled by Ratio Signal and Steady-State Heating of Time-Domain Thermoreflectance

Zhe Cheng, Fengwen Mu,* Xiaoyang Ji, Tiangui You, Wenhui Xu, Tadatomo Suga, Xin Ou, David G. Cahill, and Samuel Graham*

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ABSTRACT: Thermal resistances from interfaces impede heat dissipation in micro/nanoscale electronics, especially for high-power electronics. Despite the growing importance of understanding interfacial thermal transport, advanced thermal characterization techniques that can visualize thermal conductance across buried interfaces, especially for nonmetal−nonmetal interfaces, are still under development. This work reports a dual-modulation-frequency time-domain thermoreflectance (TDTR) mapping technique (1.61 and 9.3 MHz) to visualize the thermal conduction across buried semiconductor interfaces for β-Ga_2O_3-SiC samples. Both the β-Ga_2O_3 thermal conductivity and the buried β-Ga_2O_3-SiC thermal boundary conductance (TBC) are visualized for an area of 200 × 200 μm simultaneously. Areas with low TBC values ($\leq$20 MW/m^2·K) are identified on the TBC map, which correspond to weakly bonded interfaces caused by high-temperature annealing. Additionally, the steady-state temperature rise induced by the TDTR laser, usually ignored in TDTR analysis, is found to be able to probe TBC variations of the buried interfaces without the typical limit of thermal penetration depth. This technique can be applied to detect defects/voids in deeply buried heterogeneous interfaces nondestructively and also opens a door for the visualization of thermal conductance in nanoscale nonhomogeneous structures.

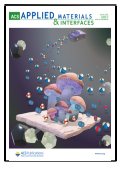
KEYWORDS: *TDTR, gallium oxide, steady-state heating, thermal visualization, thermal management, thermal boundary conductance*

INTRODUCTION

With the minimization of microelectronics and the increasing power and frequency of power and radiofrequency (RF) electronics, proper thermal management in these devices is essential for a stable and reliable performance.[1] Thermal resistances derived from interfaces in these systems show a growing importance and account for a large or even dominant part of the total thermal resistances.[2−4] Experimental measurements and theoretical predictions of interfacial thermal resistances have made great progress in the past several decades.[5−9] Different theoretical models and methods have been built to understand experimentally measured thermal boundary conductance (TBC, the inverse of thermal boundary resistance), such as the acoustic mismatch model, diffuse mismatch model, atomic Green's function, and molecular dynamics simulations.[2,6] However, advanced thermal characterization techniques to probe thermal conduction in heterogeneous structures nondestructively are under development, especially for buried structures.[10−17] Ultrafast laser based pump-probe techniques, such as time-domain thermoreflectance (TDTR) and frequency-domain thermoreflectance (FDTR), have been used to map inhomogeneous samples and study the relation between local thermal conductivity variation and microscale structural features such as grain boundaries, composition variations, and interface quality.[10,11,13,14]

Despite the great importance of understanding the thermal transport across heterogeneous interfaces, the mapping of

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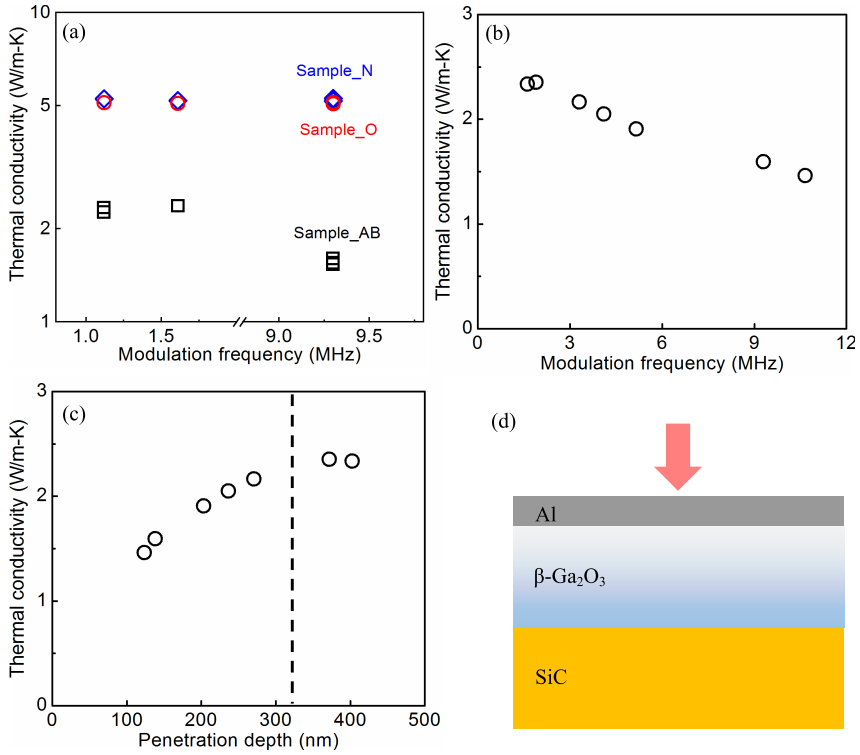


Figure 1. (a) Thermal conductivity comparison of Sample_N, Sample_O, and Sample_AB. (b) Frequency-dependent thermal conductivity of Sample_AB. (c) Penetration-depth-dependent thermal conductivity of Sample_AB. (d) Schematic diagram of the cross section of Sample_AB.

thermal conductance has not been reported previously for nonmetal−nonmetal interfaces. It is partly due to the small sensitivity resulting from the limited thermal penetration depth and the difficulty to separate multiple unknown parameters. Here, thermal penetration depth (d_p) is an estimate of how deep the periodical heating at the sample surface penetrates into the sample that depends on the modulated heating frequency (f) and thermal diffusivity (α) of the sample, namely, $d_p = \sqrt{\alpha/\pi f}$.

TDTR measures the transient thermal response of laser heating on the sample surface by periodically heating the sample surface (pump beam) and monitoring its temperature variation via thermoreflectance (probe beam).[18] By taking advantage of the transient response, the TDTR ratio signal excludes the possible errors from the variations from laser power and thermoreflectance coefficient of the transducer. However, the modulation frequency of the pump beam limits the thermal penetration depth. Only the part of the sample near the sample surface can be probed, leading to the small sensitivity of measuring deeply buried interfaces. Advanced thermal characterization techniques are on demand to break this limit and achieve high sensitivity of the TBC of the buried interfaces.[19]

The visualization of buried interfaces not only contributes to the fundamental understanding of interfacial thermal transport but also facilitates the defect detection in semiconductor devices.[17] For the range of several hundreds of nanometers to a few micrometers, this technique possibly can be a very good addition to confocal scanning acoustic microscopy that is widely used for interface quality detection in the semiconductor industry but cannot reach the thickness range mentioned above.[20]

In this work, we develop a double-frequency TDTR mapping technique to visualize the thermal conductivity of β-Ga_2O_3 thin films and the thermal conductance across buried β-Ga_2O_3-SiC interfaces. By taking advantage of the different sensitivities to unknown parameters with different modulation frequencies, two measurements with two modulation frequencies (1.61 and 9.3 MHz) are performed at the same spot to determine the unknown parameters simultaneously.[21,22] A measurement with a high modulation frequency (9.3 MHz) is performed to determine the thermal conductivity of the monocrystalline β-Ga_2O_3 layer, while a measurement with a low modulation frequency (1.61 MHz) is performed at the same spot to measure the β-Ga_2O_3-SiC TBC. Moreover, to extend the thermal penetration depth limit of thermal measurements, we utilize the steady-state heating (SSH) data of TDTR that are usually ignored in the applications of TDTR. We show that SSH can be used to infer the structural inhomogeneity of buried interfaces without the limit of the typical thermal penetration depth of TDTR.

RESULTS AND DISCUSSIONS

Three bonded β-Ga_2O_3-SiC samples are studied in this work. The monocrystalline β-Ga_2O_3 layers on 4H-SiC were fabricated by bonding with SiC by a surface activated bonding technique[23] and subsequently exfoliated from a bulk crystal by ion cutting.[24] The details about the sample fabrication can be found in the Experimental Section. The three samples are ~300-nm-thick β-Ga_2O_3 layers bonded with SiC substrates. Sample_N was annealed in N_2 at 800 °C for 30 min, while Sample_O was annealed in O_2 at 800 °C. Sample_AB was tested as bonded without annealing. An ~80-nm-thick Al was coated on the sample surface as the TDTR transducer. The thermal conductivity of the (0001)-oriented semi-insulating 4H-SiC substrate was measured from the backside of the

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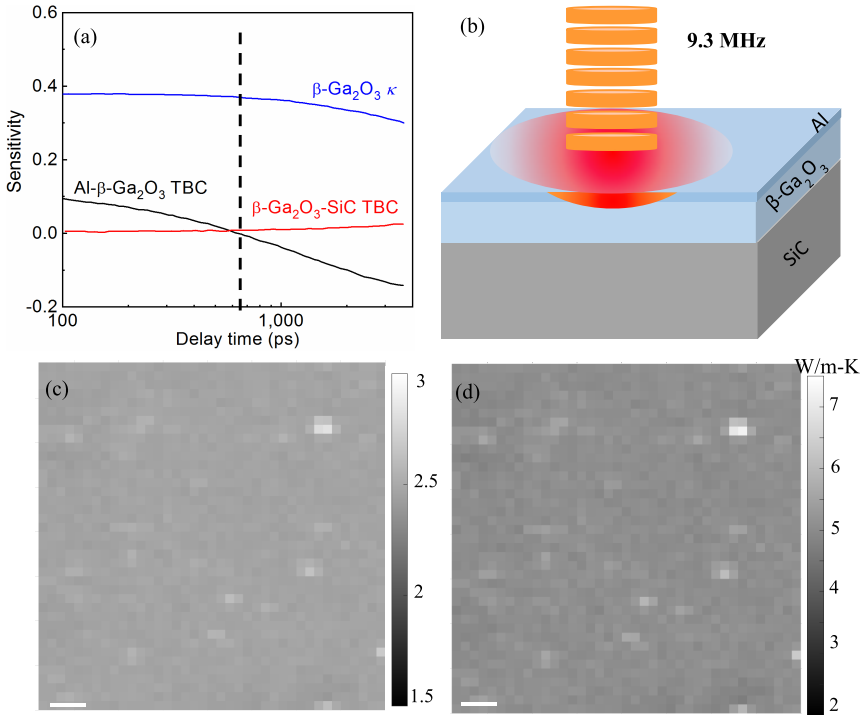


Figure 2. (a) TDTR sensitivity of Al-β-Ga_2O_3 TBC, β-Ga_2O_3 κ, and β-Ga_2O_3-SiC TBC with a modulation frequency of 9.3 MHz on Sample_O. The values of these three parameters used for sensitivity calculation are 75 MW/m^2·K, 5.1 W/(m·K), and 110 MW/m^2·K, respectively. The dash line shows that the sensitivity of the Al-β-Ga_2O_3 TBC is zero at a delay time of 650 ps. (b) Schematic diagram of TDTR measurements with a modulation frequency of 9.3 MHz. A high modulation frequency results in a short thermal penetration depth and low sensitivity of the buried β-Ga_2O_3-SiC TBC. (c) The TDTR ratio map of Sample_O. The scale bar is 20 μm, and the mapped area is 200 × 200 μm. (d) The β-Ga_2O_3 κ map of Sample_O. The scale bar is 20 μm.

sample to be 381 W/(m·K), which is consistent with literature values[25] and was used as the SiC thermal conductivity in the data fitting of other measurements.

The measured thermal conductivity of the β-Ga_2O_3 layers in Sample_N, Sample_O, and Sample_AB is shown in Figure 1a. The two annealed samples have similar thermal conductivities and are more than twice as large as the as-bonded one. The annealing removes the ion-implantation-induced strain and restores the lattice distortion in the β-Ga_2O_3 layers, resulting in the increase in thermal conductivity, similar to our previous work.[24] The β-Ga_2O_3 thermal conductivity of Sample_N and Sample_O does not show any dependence on modulation frequency, showing that the β-Ga_2O_3 defect/vacancy concentrations are relatively uniform. The strong modulation frequency dependence of the measured β-Ga_2O_3 thermal conductivity of Sample_AB is shown in Figure 1b. Each modulation frequency corresponds to a thermal penetration depth, as shown in Figure 1c. The β-Ga_2O_3 film thickness of Sample_AB is marked as a dash line. When the thermal penetration depth is larger than the film thickness, the laser heating penetrates through the β-Ga_2O_3 film. The measured β-Ga_2O_3 thermal conductivity keeps constant.

The penetration-depth-dependent thermal conductivity shows that Sample_AB has a gradient defect/vacancy concentration distribution, as shown in Figure 1d. When ions implant a bulk β-Ga_2O_3 wafer, ions travel through the top surface and accumulate in a layer several hundreds of nanometers beneath the surface. The damage caused by ion implantation has a gradient from the top surface (low concentration) to the accumulated layer (high concentration). Then, the top surface of the wafer is bonded with a SiC wafer. After heating up the β-Ga_2O_3-SiC wafer composite, the thermal stresses cause the sample to fracture along the defect plane created by the implanted ions, which separates the several hundreds of nanometers of β-Ga_2O_3 from the β-Ga_2O_3 wafer. The exposed surface of the β-Ga_2O_3 thin film is the sample surface in this work and also the exfoliation interface. As shown in Figure 1d, the top part of the β-Ga_2O_3 layer is close to the exfoliation interface where a large amount of implanted hydrogen ions is accumulated to break the β-Ga_2O_3 thin layer from the β-Ga_2O_3 bulk crystal, which has higher concentrations of ion-induced disorder than the part of β-Ga_2O_3 close to the SiC wafer. Higher concentrations of disorder result in lower thermal conductivity.

In TDTR measurements, the sensitivity of measuring a specific unknown parameter is defined as follows:

$$S_i = \frac{\partial \ln(-V_{in}/V_{out})}{\partial \ln(p_i)}, \quad (1)$$

where S_i is the sensitivity to parameter i, $-V_{in}/V_{out}$ is the TDTR signal, and p_i is the value of parameter i. For the sample structures in this work, there are three unknown parameters: Al-β-Ga_2O_3 TBC, β-Ga_2O_3 thermal conductivity (κ), and β-Ga_2O_3-SiC TBC. As shown in Figure 2a, the calculated sensitivity of each parameter changes with the delay time of TDTR. The dash line shows that the sensitivity of the Al-β-Ga_2O_3 TBC is zero at a delay time of 650 ps with a modulation frequency of 9.3 MHz on Sample_O. At this delay time, the measured TDTR ratio is not affected by the Al-β-Ga_2O_3 TBC. The schematic diagram of the transient heating of the sample surface with a modulation frequency of 9.3 MHz is shown in Figure 2b. The thermal penetration depth is smaller than the

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film thickness. Therefore, TDTR is sensitive to the β-Ga_2O_3 κ, but the sensitivity of the β-Ga_2O_3-SiC TBC is almost zero. At a delay time of 650 ps, the TDTR ratio is only related to one unknown parameter (β-Ga_2O_3 κ).

Figure 2c shows a 200 × 200 μm map of the TDTR ratio. The step size of the mapping is 5 μm. A 10× objective was used with a laser spot size of 5.5 μm in radius. We typically argue that the spatial resolution is $1/\sqrt{2}$ of the spot size, although in practice, the resolution is worse than that because of the lateral heat spreading.[10] For our case in this work, the typically argued spatial resolution ($5.5/\sqrt{2} = 3.89$ μm) is smaller than the step size of the mapping. Therefore, we take the step size (5 μm) as the spatial resolution of this mapping. We choose the step size of 5 μm for the mapping experiments by considering both the laser spot size and the size of the weakly bonded area. The small step size (<5 μm) results in the overlapping of two measurements. Moreover, a 5-μm step size is enough to detect the relative TBC variation of the buried interface. The dwell time for each step is 0.5 s, and the time traveling from the last spot to the first spot on the next line is 1 s. All the mapping parameters mentioned here are the same for the following mappings. The β-Ga_2O_3 κ distribution is shown in Figure 2d. Most of the spots have a thermal conductivity close to 5 W/(m·K), which is close to the measured value of the full-range TDTR measurements in Figure 1a.

Unlike the high modulation frequency used to measure the β-Ga_2O_3 κ in Figure 2a, Figure 3a shows the sensitivity of Al-β-Ga_2O_3 TBC, β-Ga_2O_3 κ, and β-Ga_2O_3-SiC TBC with a modulation frequency of 1.61 MHz on Sample_O. The measurements with low modulation frequency are performed at the same area as the 9.3 MHz measurements. The low modulation frequency results in a large thermal penetration depth. TDTR has enough sensitivity to probe the Ga_2O_3-SiC TBC. At 330 ps as shown by the dash line in Figure 3a, the sensitivity of the Al-β-Ga_2O_3 TBC is zero. The measured TDTR ratio is only related to β-Ga_2O_3 κ and β-Ga_2O_3-SiC TBC. The β-Ga_2O_3 κ at each spot has been measured with the 9.3 MHz modulation frequency as discussed above. The measured TDTR ratio with a modulation frequency of 1.61 MHz is shown in Figure 3b. It is obvious that there are some locations with very low TDTR ratios. The β-Ga_2O_3 κ map has been obtained from the high-frequency measurements. By fitting the TDTR ratio of the 1.61 MHz measurements and inserting the β-Ga_2O_3 κ values as known parameters, the β-Ga_2O_3-SiC TBC at each location can be extracted, as shown in Figure 3c. By calculating all the spots in the mapping area, we obtain the TBC map. The low TDTR ratio corresponds to a low β-Ga_2O_3-SiC TBC. The measured β-Ga_2O_3-SiC TBC is as low as 20 MW/m^2·K, which is close to the TBC values of van der Waals bonded interfaces.[26] These areas should have very weak bonds between β-Ga_2O_3 and SiC substrates. A red dash circle and a blue dash circle are used to mark a typical well-bonded area and a weakly bonded area. Full-range TDTR scans will be performed on these spots to check the mapping results and presented later in this work. The visualization capability shown here can be applied to detect defects in buried semiconductor structures.

Figure 4a shows the statistical distribution of the mapped β-Ga_2O_3 thermal conductivity. The measured thermal conductivity is quite uniform. More than 75% of the mapped spots have a thermal conductivity between 5 and 5.25 W/(m·K). About 95% of the mapped spots have a thermal conductivity

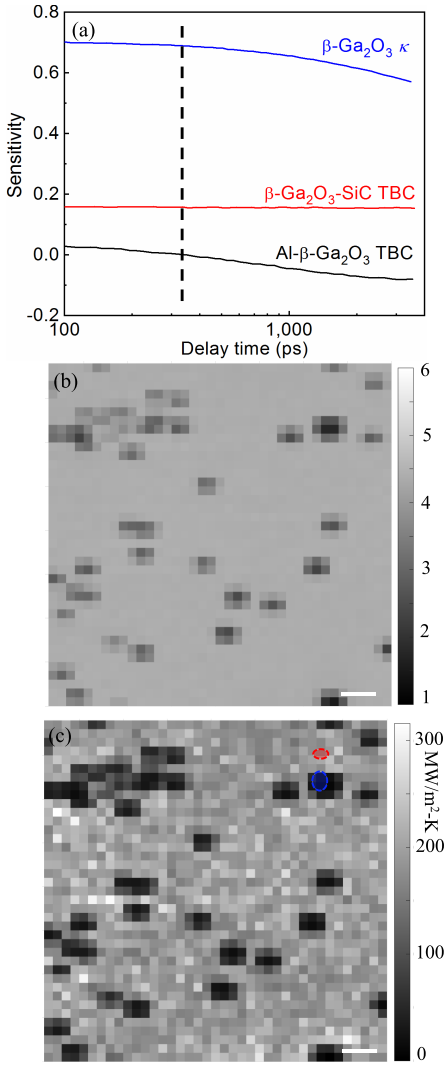


Figure 3. (a) TDTR sensitivity of Al-β-Ga_2O_3 TBC, β-Ga_2O_3 κ, and β-Ga_2O_3-SiC TBC with a modulation frequency of 1.61 MHz on Sample_O. The values of these three parameters used for sensitivity calculation are 75 MW/m^2·K, 5.1 W/(m·K), and 110 MW/m^2·K, respectively. The dash line shows that the sensitivity of Al-β-Ga_2O_3 TBC is zero at a delay time of 330 ps. (b) The TDTR ratio map of Sample_O. The scale bar is 20 μm, and the mapped area is 200 × 200 μm. (c) The β-Ga_2O_3-SiC TBC map of Sample_O. The scale bar is 20 μm. The red and blue dash circles are used to mark a typical well-bonded and weakly bonded spots.

between 4.75 and 5.25 W/(m·K). This is consistent with the results measured by the full-range TDTR scans, as shown in Figure 1a. Figure 4b shows the statistical distribution of the mapped β-Ga_2O_3-SiC TBC. The measured TBC values spread over a wide range, indicating the nonuniform quality of the buried β-Ga_2O_3-SiC interfaces. There are a large number of spots that have a TBC lower than 50 MW/m^2·K. The sensitivity is small (<0.1) for high TBC values (>150 MW/m^2·K). Here, we only focus on the areas with low TBC values. These spots are weakly bonded and lead to heat dissipation challenges, which need to be identified in the buried semiconductor interfaces.

To double check our mapped results, a typical well-bonded area and a typical weakly bonded area are chosen to perform full-range TDTR scans, as marked by red and blue dash circles in Figure 3c. Figure 4c shows the acoustic echoes of the Al-β-

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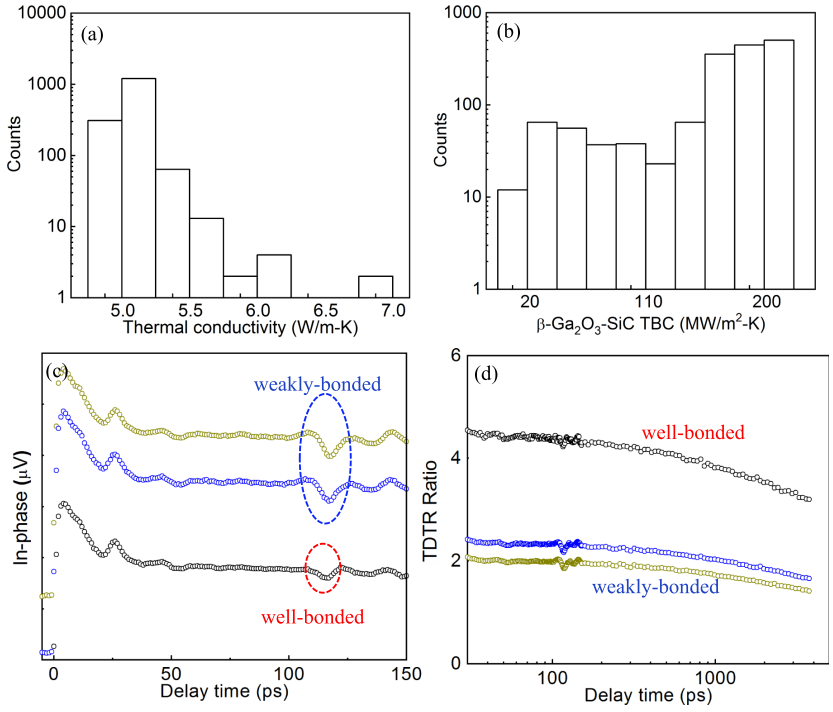


Figure 4. (a) Statistical distribution of the mapped β-Ga_2O_3 κ. (b) Statistical distribution of the mapped β-Ga_2O_3-SiC TBC. (c) Acoustic echoes of the Al-β-Ga_2O_3 interfaces and the β-Ga_2O_3-SiC interfaces measured by the picosecond acoustic technique on the selected areas as marked by red and blue circles in Figure 3c. (d) TDTR ratio curves of measurements on the selected areas as marked by red and blue circles in Figure 3c.

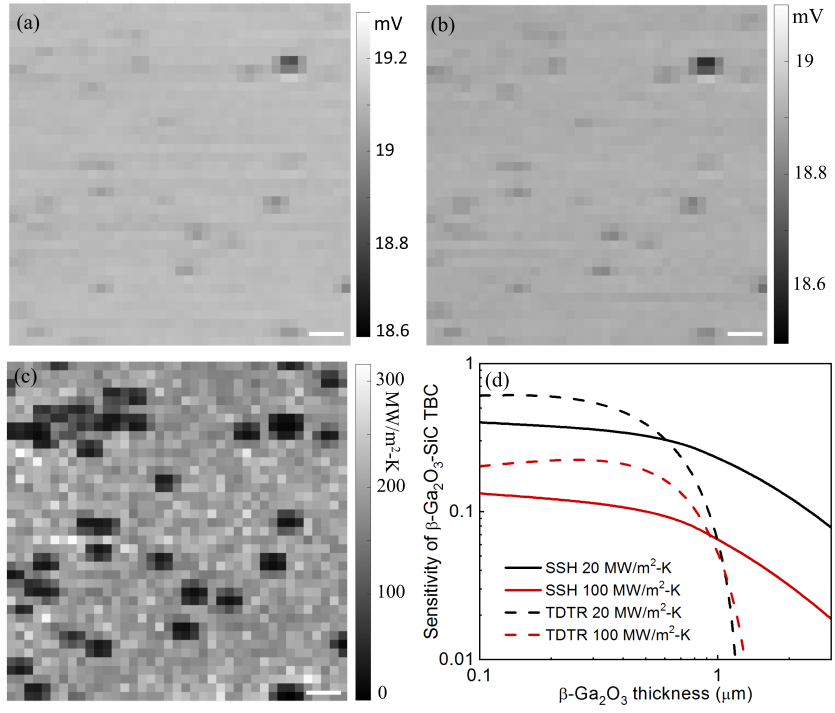


Figure 5. (a) The detector voltage of Sample_O with a modulation frequency of 9.3 MHz. (b) The detector voltage of Sample_O with a modulation frequency of 1.61 MHz. (c) The β-Ga_2O_3-SiC TBC is included for comparison. (d) Sensitivity of β-Ga_2O_3-SiC TBC for TDTR measurements and steady-state-heating (SSH) method as a function of β-Ga_2O_3 thickness. The modulation frequency of TDTR is 1.61 THz.

Ga_2O_3 interfaces and the β-Ga_2O_3-SiC interfaces measured by the picosecond acoustic technique on the selected areas. The echoes at a delay time of ~25 ps are from the Al-β-Ga_2O_3 interfaces, while the echoes at a delay time of ~118 ps are from the β-Ga_2O_3-SiC interfaces. The echoes of the Al-β-Ga_2O_3 interfaces for all three measurements are the same. However, the echoes from the spots with weakly bonded β-Ga_2O_3-SiC interfaces are much stronger than the echo of the spot with a well-bonded β-Ga_2O_3-SiC interface. The echoes confirm the structural difference of the β-Ga_2O_3-SiC interfaces between these spots.

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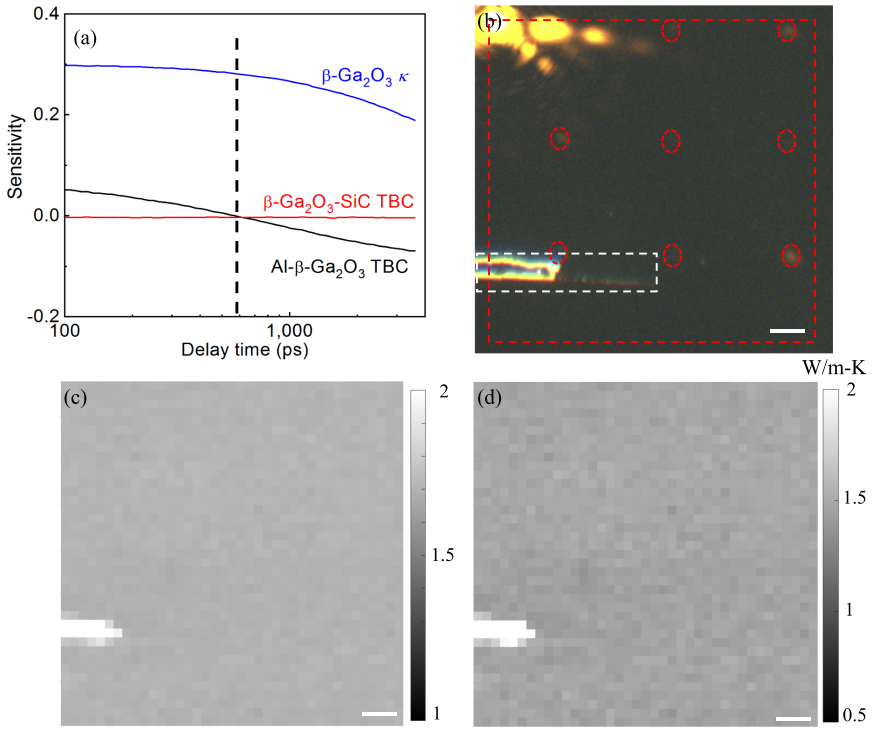


Figure 6. (a) TDTR sensitivity of Al-β-Ga_2O_3 TBC, β-Ga_2O_3 κ, and β-Ga_2O_3-SiC TBC with a modulation frequency of 9.3 MHz on Sample_AB. The dash line shows that the sensitivity of the Al-β-Ga_2O_3 TBC is zero at a delay time of 580 ps. (b) Sample surface captured by a CCD camera. The red dash line square marks the mapped area. A surface defect is included as a marker in the measurements for comparison. The TDTR laser spot shoots on the upper left corner of the red square. The scale bar is 20 μm, and the mapped area is 200 × 200 μm. (c) The TDTR ratio map of Sample_AB. The scale bar is 20 μm. (d) The β-Ga_2O_3 κ map of Sample_AB. The scale bar is 20 μm.

Figure 4d shows the full-range TDTR ratios measured on the selected areas. The ratio of the measurement on the well-bonded spot is much larger than that of the weakly bonded spots. This also confirms the difference of thermal properties of these areas. By fitting the full-range TDTR ratios, the fitted β-Ga_2O_3-SiC TBC is about 10 MW/m^2·K for the weakly bonded β-Ga_2O_3-SiC interfaces and about 160 MW/m^2·K for the well-bonded β-Ga_2O_3-SiC interfaces, which are close to the mapped TBC values.

Figure 5a,b shows the detector voltage of Sample_O with modulation frequencies of 9.3 and 1.61 MHz. The detector voltage measures the power of the reflected probe beam from the sample surface. If we assume that the laser power does not change, the variation of the detector voltage during the mapping is proportional to the variation of the reflectivity. The ratio of the change in reflectivity and the change in temperature is approximately a constant (thermoreflectance coefficient). A typical value of the thermoreflectance coefficient of Al is 2.1 × 10^{-4} K^{-1} at 785 nm wavelength.[27] The variation of reflectivity is proportional to the variation of surface temperature. The maps in Figure 5a,b show that the steady-state temperature rises due to the TDTR laser heating with the laser power drift corrected in the data processing. More details about the data processing can be found in the Supporting Information. Previous TDTR measurements usually ignore this steady-state heating data and the information that can be extracted. The relative variation of the surface reflectivity shows the relative variation of the surface temperature on different spots and corresponding thermal resistances from the sample surface to heat sinks on different spots.

The TBC map is also included in Figure 5c as comparison. These TDTR measurements with a high modulation frequency have no sensitivity to the buried interfaces, while the steady-state heating captures the information of the buried interfaces. Figure 5d shows the sensitivity of β-Ga_2O_3-SiC TBC for TDTR measurements and steady-state heating (SSH) method as a function of β-Ga_2O_3 thickness. The modulation frequency of TDTR is 1.61 THz. The sensitivity of the SSH method is calculated similar to TDTR by replacing the TDTR ratio as the steady-state temperature rises. A small β-Ga_2O_3-SiC TBC overall has high sensitivity since the thermal resistance from the interface is relatively large compared with the thermal resistances of other layers. As the β-Ga_2O_3 thickness increases, the sensitivity of β-Ga_2O_3-SiC TBC decreases because the heat needs to penetrate through the whole β-Ga_2O_3 layer to reach the interface. For thin films (<600 nm), TDTR has a higher sensitivity than SSH. However, SSH can probe much more deeply than TDTR for buried high thermally resistive defects such as weakly bonded interfaces.

If we set a sensitivity of 0.1 as the cutoff point, for the weakly bonded area (20 MW/m^2·K), the β-Ga_2O_3 thickness limit for TDTR measurements is ~0.9 μm, while that for SSH is ~2.45 μm. For the mapped area in this work with high β-Ga_2O_3-SiC TBC (>100 MW/m^2·K), the sensitivity is low for both TDTR and SSH. From a practical point, the weakly bonded area is the one the measurements are detecting. The SSH method is favorable to detect deeply buried defects with large thermal resistance.

Sample_AB is also mapped with a modulation frequency of 9.3 MHz. Sample_AB has a much lower thermal conductivity than Sample_N and Sample_O, which reduces the thermal penetration depth. TDTR measurements do not have enough sensitivity to the buried β-Ga_2O_3-SiC TBC even with a low modulation frequency. Therefore, the steady-state heating data

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are used to infer the thermal properties of the interfaces among different spots. As shown in Figure 6a, the sensitivity of the Al-β-Ga_2O_3 TBC is zero at the delay time of 580 ps. The sensitivity of the buried β-Ga_2O_3-SiC TBC is close to zero for all the delay times. Therefore, the TDTR ratio is only related to the β-Ga_2O_3 κ. As shown in Figure 6b captured by a charge-coupled device (CCD) camera, an area with some defects on the surface is chosen for the mapping, marked by the red dash square. The defect area is marked by the white dash rectangle. The spot in the upper left corner is from the TDTR laser. When the laser beam is focused on the surface, the apparent array of secondary spots marked by red dash circles shows up, which is a known artifact of our CCD camera. These spots are not from the reflections of surface defects on the sample.[28]

Figure 6c shows the mapped ratio of Sample_AB. The measured TDTR ratio is very uniform except for the defective area. The TDTR ratio is fitted to an analytical heat transfer solution of the sample structure to obtain the β-Ga_2O_3 κ. The mapped β-Ga_2O_3 κ is shown in Figure 6d. The uniform κ map indicates the uniform quality of the β-Ga_2O_3 thin films.

The detector voltage of the mapping on Sample_AB with a modulation frequency of 9.3 MHz is shown in Figure 7. The

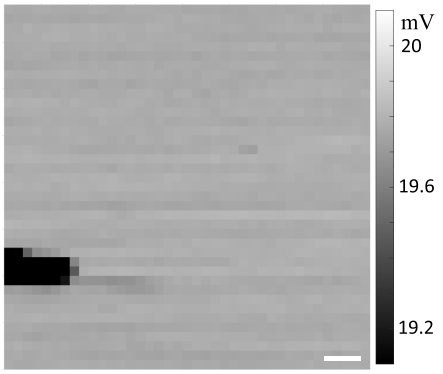


Figure 7. The detector voltage map of Sample_AB with a modulation frequency of 9.3 MHz. The scale bar is 20 μm, and the mapped area is 200 × 200 μm.

uniform detector voltage distribution shows uniform distributed temperature rises and corresponding uniform distributed thermal resistances, unlike those of Sample_O. According to the above discussion about the steady-state heating of TDTR measurements, we speculate that the buried β-Ga_2O_3-SiC TBC distribution is uniform. The difference of sample fabrication between Sample_O and Sample_AB is annealing. Sample_O is annealed at 800 °C in O_2, while Sample_AB is as-bonded without annealing. The high-temperature annealing improves the quality of the β-Ga_2O_3 layer but induces the nonuniformity in the buried interfaces possibly due to the local stress.[23]

In terms of error analysis, for a typical TDTR measurement, the uncertainty of the measured thermal conductivity is about ±8%. For the TBC of the buried β-Ga_2O_3-SiC interfaces, the error bar that is calculated based on a Monte Carlo method depends on the sensitivity of the TBC. For a Ga_2O_3-SiC TBC value of about 20 MW/m^2·K, the typical error bar is about ±12%.[26] For a Ga_2O_3-SiC TBC value of about 100 MW/m^2·K, the typical error bar is about ±30%.[24] Since we did the mapping on the same sample, the systematic errors should be similar for different spots. When mapping the relative TBC variation of the buried interface, this mapping technique should have smaller error bars than the typical values mentioned above.

Quasi-ballistic thermal transport may result in different measured thermal conductivities with different laser spot sizes and frequencies in TDTR measurements.[29−31] Our samples in this work are not expected to have such effect. Most of the phonons in β-Ga_2O_3 have mean free paths smaller than 1 μm, much smaller than our laser spot size.[32,33] We did not see a frequency-dependent thermal conductivity for the annealed samples, as shown in Figure 1a.

CONCLUSIONS

This work reports a dual-modulation-frequency TDTR mapping technique to visualize the thermal conductance across buried semiconductor interfaces. The mapped β-Ga_2O_3 thermal conductivity is uniform (about 5 W/(m·K) for a sample annealed at 800 °C in O_2). We observe some areas with very low TBC values (≤20 MW/m^2·K) on the TBC map, which correspond to weakly bonded interfaces caused by high-temperature annealing. The steady-state heating data in the TDTR measurements are found to be amenable to probing the TBC variation of buried interfaces. The temperature distribution map obtained from the reflectivity map and detector voltage map matches well with the measured TBC map of the buried interfaces. After mapping an as-bonded β-Ga_2O_3-SiC sample with a high modulation frequency of 9.3 MHz, both a uniform thermal conductivity map and a uniform detector voltage map are obtained in the in-plane direction. The TBC maps of the buried β-Ga_2O_3-SiC interfaces are also uniform. Our work is able to detect defects/voids in deeply buried heterogeneous interfaces nondestructively and enables the visualization of thermal conduction in nanoscale non-homogeneous structures.

EXPERIMENTAL SECTION

Sample Preparation. The 4-in. on-axis (0001) semi-insulating 4H-SiC wafers and the 2-in. ($\bar{2}$01) β-Ga_2O_3 wafers were purchased from SICC Co., Ltd., and Novel Crystal Technology Inc. The surfaces were polished by chemical mechanical polishing (CMP) to root-mean-square surface roughness values of ∼0.3 and ∼0.27 nm for SiC and β-Ga_2O_3, respectively. The β-Ga_2O_3 wafers were implanted by hydrogen ions with an energy of 35 keV and an implantation dose of ∼1 × 10^{17} cm^{-2} at room temperature with a 7° tilt angle. The hydrogen ions accumulate in the β-Ga_2O_3 at 200−400 nm beneath the surface.

The β-Ga_2O_3 is then bonded with the SiC wafer by surface-activated bonding at room temperature. The background pressure is kept at 5 × 10^{-6} Pa. A silicon source is added into the Ar source to ion-implant both the SiC and β-Ga_2O_3 surfaces with a power of 1.0 kV and 100 mA. The Ar ions remove any surface contaminations and activate the SiC and β-Ga_2O_3 surfaces before the wafers are put in contact for bonding. The chamber pressure is about 0.4 Pa. More details about the bonding and ion implantation can be found in the literature.[24,34]

After that, the implanted hydrogen ions induce the exfoliation of a β-Ga_2O_3 thin film after heating the bonded wafers at 450 °C. The exfoliated β-Ga_2O_3 film is polished by CMP to a thickness of ∼300 nm. Sample_AB is the as-bonded sample, while Sample_N and Sample_O are annealed at 800 °C in N_2 and O_2 for 30 min, respectively. For TDTR measurements, a layer of ∼80-nm-thick Al was deposited by DC (direct current) sputtering on the surface of all samples. The thermal conductivity of the Al film is ∼170 W/(m·K) by measuring its electrical conductivity and applying the Wiedemann−Franz law.

TDTR Measurements. The TDTR system used in this work is a two-tint pump-probe system.[35] A mode-locked femtosecond Ti-

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sapphire laser is split into a pump beam and a probe beam. The pump beam modulated by an electro-optic modulator heats the sample surface periodically, while the probe beam detects the temperature variation of the sample surface via thermoreflectance.[18] The signal picked up by a photodetector and a lock-in amplifier is fitted with an analytical heat transfer solution of the sample structure to infer unknown thermal properties.[36] A 10× objective is used in this work with beam sizes of 5.5 μm in radius. The local thicknesses of Al and β-Ga_2O_3 films were determined by the picosecond acoustic technique.[26] The sound velocity of β-Ga_2O_3 along the cross-plane direction is 6.6 km/s.[26,37] The heat capacity data are from the literature.[25,38]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.1c06212.

The supporting information includes the data processing of correcting the detector voltage with a linear laser power decaying with time (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Fengwen Mu − *Kagami Memorial Research Institute for Materials Science and Technology, Waseda University, Tokyo 169-0051, Japan;* orcid.org/0000-0001-7396-9347; Email: mufengwen123@gmail.com

Samuel Graham − *George W. Woodruff School of Mechanical Engineering, Georgia Institute of Technology, Atlanta, Georgia 30332, United States;* orcid.org/0000-0002-1299-1636; Email: sgraham@gatech.edu

Authors

Zhe Cheng − *Department of Materials Science and Engineering and Materials Research Laboratory, University of Illinois at Urbana-Champaign, Urbana, Illinois 61801, United States;* orcid.org/0000-0001-7827-2979

Xiaoyang Ji − *Department of Materials Science and Engineering and Materials Research Laboratory, University of Illinois at Urbana-Champaign, Urbana, Illinois 61801, United States*

Tiangui You − *State Key Laboratory of Functional Materials for Informatics, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai 200050, China*

Wenhui Xu − *State Key Laboratory of Functional Materials for Informatics, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai 200050, China*

Tadatomo Suga − *Collaborative Research Center, Meisei University, Tokyo 191-8506, Japan*

Xin Ou − *State Key Laboratory of Functional Materials for Informatics, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai 200050, China;* orcid.org/0000-0002-0316-9958

David G. Cahill − *Department of Materials Science and Engineering and Materials Research Laboratory, University of Illinois at Urbana-Champaign, Urbana, Illinois 61801, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsami.1c06212

Notes

The authors declare no competing financial interest.

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